

PL/SQL Exercise 3: Stored Procedures

Scenario 1: The bank needs to process monthly interest for all savings accounts.

- **Question:** Write a stored procedure **ProcessMonthlyInterest** that calculates and updates the balance of all savings accounts by applying an interest rate of 1% to the current balance.

Accounts Table:

	ACCOUNTID	CUSTOMERNAME	ACCOUNTTYPE	BALANCE
1	101	Rahul	Savings	5000
2	102	Ananya	Savings	12000
3	103	Vikram	Current	25000
4	104	Priya	Savings	8000
5	105	Kiran	Current	15000

Solution Query Output:

EXECUTE ProcessMonthlyInterest

	ACCOUNTID	CUSTOMERNAME	ACCOUNTTYPE	BALANCE
1	101	Rahul	Savings	5050
2	102	Ananya	Savings	12120
3	103	Vikram	Current	25000
4	104	Priya	Savings	8080
5	105	Kiran	Current	15000

Scenario 2: The bank wants to implement a bonus scheme for employees based on their performance.

- **Question:** Write a stored procedure **UpdateEmployeeBonus** that updates the salary of employees in a given department by adding a bonus percentage passed as a parameter.

Employees Table:

	EMPLOYEEID	EMPLOYEEENAME	DEPARTMENT	SALARY
1	101	Rahul	IT	50000
2	102	Ananya	IT	60000
3	103	Vikram	HR	45000
4	104	Priya	HR	55000
5	105	Kiran	Finance	70000

Solution Query Output:

EXECUTE UpdateEmployeeBonus(10)

	EMPLOYEEID	EMPLOYEEENAME	DEPARTMENT	SALARY
1	101	Rahul	IT	55000
2	102	Ananya	IT	66000
3	103	Vikram	HR	45000
4	104	Priya	HR	55000
5	105	Kiran	Finance	70000

Scenario 3: Customers should be able to transfer funds between their accounts.

- **Question:** Write a stored procedure **TransferFunds** that transfers a specified amount from one account to another, checking that the source account has sufficient balance before making the transfer.

Accounts Table:

	ACCOUNTID	CUSTOMERNAME	ACCOUNTTYPE	BALANCE
1	101	Rahul	Savings	5000
2	102	Ananya	Savings	12000
3	103	Vikram	Current	25000
4	104	Priya	Savings	8000
5	105	Kiran	Current	15000

Solution Query Output:

EXECUTE TransferFunds(101,102,1000)

	ACCOUNTID	CUSTOMERNAME	ACCOUNTTYPE	BALANCE
1	101	Rahul	Savings	4000
2	102	Ananya	Savings	13000
3	103	Vikram	Current	25000
4	104	Priya	Savings	8000
5	105	Kiran	Current	15000